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MINI PROJECT REPORT

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English Inflectional and Derivational Morphology

1. Project Title

English Inflectional and Derivational Morphology

2. Introduction

Morphology is the branch of linguistics and Natural Language Processing (NLP) that studies the internal structure of words and how words are formed from smaller meaningful units called morphemes. English words are commonly built through two major morphological processes: inflection and derivation.

Inflectional morphology modifies a word to express grammatical features such as tense, number, person, or degree of comparison, without changing its part of speech or core meaning (for example, walk to walked, cat to cats, tall to taller). Derivational morphology, on the other hand, creates new words, often with a different part of speech or a modified meaning, by attaching prefixes or suffixes to a root or base (for example, happy to happiness, act to reactivate, care to careless).

In this project, a rule-based morphological analysis system is developed and evaluated on a dataset of English words. The primary objective is to automatically identify whether a word is formed through inflection or derivation, extract the root/base form and the affix, and determine the accuracy of automated parsing across the two morpheme categories.

The analysis uses information about word forms, root words, affixes, part-of-speech tags, formation processes, word length, syllable count, and whether the affix changes the part of speech of the base word.

A statistical significance test is performed to determine whether the difference in automated parsing accuracy between inflectional and derivational words is meaningful or simply due to chance. The results are then converted into linguistic insights and practical recommendations for NLP system design.

3. Problem Statement

NLP systems such as search engines, spell-checkers, machine translators, and text-to-speech engines rely on morphological analysis to reduce words to their base forms and to understand grammatical relationships between words.

While inflectional morphology in English follows a relatively small and regular set of rules, derivational morphology is far more irregular, involves a much larger set of affixes, and frequently changes the part of speech and meaning of the base word. As a result, a single rule-based parser may not perform equally well on both morpheme types.

Therefore, an NLP practitioner needs an analytical approach to:

- Compare the composition of inflectional and derivational word forms in an English text corpus.
- Analyze the structural and linguistic characteristics of each morpheme type.
- Test whether the difference in automated morphological-parsing accuracy is statistically significant.
- Identify which morpheme type is harder for a rule-based parser to analyze correctly.
- Provide data-driven recommendations for improving morphological analyzers.

4. Aim / Goal

The main goal of this project is to perform a computational morphological analysis of English words and determine how accurately a rule-based parser can identify and decompose inflectional versus derivational word forms.

The project aims to use statistical analysis to validate the difference in parsing accuracy between the two morpheme categories and to recommend improvements for NLP morphological-analysis pipelines.

5. Objectives

1. To collect and prepare an English word dataset for morphological analysis.
2. To categorize words as inflectional or derivational forms.
3. To calculate the parsing accuracy for both morpheme categories.
4. To analyze structural characteristics such as word length, syllable count, and affix length.
5. To examine parsing accuracy across part-of-speech categories and word-formation processes.
6. To formulate null and alternative hypotheses regarding parsing accuracy.
7. To perform statistical significance testing using the chi-square test.
8. To determine whether the observed difference in accuracy is statistically significant.
9. To identify which morpheme category is more challenging for automated analysis.
10. To provide recommendations for improving morphological-analysis systems.

6. Linguistic Context / Background

Morphological analysis is a foundational step in the NLP pipeline because it enables systems to normalize word variants to a common base form and to understand how meaning and grammatical function change through affixation.

A rule-based morphological analyzer can significantly improve downstream tasks such as information retrieval, lemmatization, part-of-speech tagging, and text classification when it correctly separates inflectional variation from derivational word formation.

For example, a system may encounter two related words:

- Inflectional form: "running" -> base word "run" + inflectional suffix "-ing" (same part of speech, grammatical variation).
- Derivational form: "runner" -> base word "run" + derivational suffix "-er" (different part of speech, new word).

Instead of assuming that a single stemming or lemmatization rule set will work equally well for all words, this project uses a rule-based morphological analyzer to test that assumption directly.

In this project, 1,000 English word forms are analyzed. The words are divided equally between the two morpheme categories.

7. Dataset Description

Data Source

The dataset was created specifically for this academic NLP project using a curated list of English words drawn from standard English vocabulary and annotated with morphological information.

It contains word-level records representing two morpheme categories. The dataset is designed to demonstrate data collection, preprocessing, exploratory analysis, statistical testing, visualization, and linguistic decision-making.

Data Size

- Total records: 1,000
- Inflectional word forms: 500
- Derivational word forms: 500
- Number of variables: 12

Variables / Features

Variable	Type	Description
word_id	Categorical	Unique word identifier
word	Categorical	Surface word form

base_word	Categorical	Root/base form of the word
affix	Categorical	Attached prefix or suffix
morpheme_type	Categorical	Inflectional or Derivational
pos_tag	Categorical	Part of speech of the surface word
formation_process	Categorical	Prefixation, Suffixation, Compounding, Zero-Derivation
word_length	Numerical	Number of characters in the word
syllable_count	Numerical	Number of syllables in the word
pos_changed	Binary	Whether the affix changed the part of speech
parsed_correctly	Binary	Whether the rule-based parser analyzed the word correctly
affix_length	Numerical	Number of characters in the affix

Target Variable

The primary target variable is:

parsed_correctly

- 0 = Incorrectly parsed
- 1 = Correctly parsed

The main explanatory variable is:

morpheme_type

- Inflectional = grammatical variation of a base word
- Derivational = new word formed from a base word

8. Data Collection

The dataset consists of 1,000 English word records, each manually or semi-automatically annotated with its morphological structure.

Each word is assigned to one of two morpheme categories:

- 500 words -> Inflectional forms (plural nouns, past-tense verbs, comparative/superlative adjectives, progressive/third-person verb forms)
- 500 words -> Derivational forms (nominalizations, adjectivalizations, verbalizations, and words formed using prefixes such as un-, re-, dis-, and suffixes such as -ness, -ment, -ize, -less)

Word structure is represented using linguistic and structural variables such as base word, affix, part-of-speech tag, formation process, word length, syllable count, and whether the affix changed the part of speech.

The dataset is stored in CSV format and analyzed using Python.

9. Data Preprocessing

Data preprocessing was performed before conducting the morphological analysis.

9.1 Data Cleaning

The dataset was checked for:

- Duplicate word entries
- Invalid or malformed affix annotations
- Incorrect data types
- Missing values
- Unusually long or short outlier words

The word_id field was used to identify individual word records. Categorical fields were checked to ensure that values such as part-of-speech tags and formation process were consistently represented.

9.2 Missing Value Handling

The dataset contains one missing value in the syllable_count column.

The missing numerical value can be handled using median imputation because the median is less affected by extreme observations than the mean. The imputed value is then used for EDA and further analysis.

9.3 Outlier Detection

Outlier detection was performed on numerical variables such as:

- Word length
- Syllable count
- Affix length

One unusually long compound-derived word was included in the raw dataset to demonstrate the outlier-detection process. A box plot and IQR-based analysis can be used to identify extreme values.

9.4 Data Transformation

Categorical and numerical variables were inspected and transformed where necessary. Binary variables were represented numerically:

- 0 = No
- 1 = Yes

Parsing accuracy was calculated using:

$$\text{Parsing Accuracy} = (\text{Number of Correctly Parsed Words} / \text{Total Words}) \times 100$$

9.5 Data Encoding / Normalization

Categorical variables such as:

- Part-of-speech tag
- Formation process
- Morpheme type

can be converted into numerical form using one-hot encoding when required for machine-learning-based morphological classification.

Normalization is not required for the main chi-square accuracy test because the test operates on categorical parsing outcomes rather than distance-based numerical features.

10. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the structure and characteristics of the dataset.

10.1 Morpheme Type Distribution

The number of words assigned to each morpheme category is examined.

- Inflectional = 500 words
- Derivational = 500 words

This confirms that both categories have equal sample sizes.

10.2 Parsing Accuracy Analysis

Morpheme Type	Words	Correctly Parsed	Accuracy
Inflectional	500	460	92.0%
Derivational	500	405	81.0%

Inflectional words show a higher parsing accuracy than derivational words.

10.3 Morphological Complexity Analysis

Average structural values show:

Metric	Inflectional	Derivational
Average Word Length	6.8 characters	8.4 characters
Average Syllable Count	2.1	2.9
Average Affix Length	2.3 characters	3.6 characters
Part-of-Speech Change Rate	4.0%	78.0%

The results indicate that derivational words are structurally longer, more complex, and far more likely to change part of speech than inflectional words.

10.4 Part-of-Speech Analysis

Parsing accuracy can be compared across:

- Noun
- Verb
- Adjective
- Adverb

This helps identify whether the effectiveness of the morphological parser differs according to the part of speech of the word.

10.5 Formation Process Analysis

Words are categorized according to their word-formation process:

- Suffixation (Inflectional)
- Suffixation (Derivational)
- Prefixation (Derivational)
- Compounding (Derivational)
- Zero-Derivation (Derivational)

This helps determine which formation processes are handled well by the rule-based parser and which require additional rules.

11. Statistical Analysis

Statistical analysis is the main component of this project.

Hypothesis

Null Hypothesis (H0)

There is no statistically significant difference in rule-based parsing accuracy between inflectional and derivational word forms.

Alternative Hypothesis (H1)

There is a statistically significant difference in rule-based parsing accuracy between inflectional and derivational word forms.

Significance Level

The significance level is selected as:

alpha = 0.05

If: $p\text{-value} < 0.05$, the null hypothesis is rejected.

Parsing Accuracy

For Inflectional words: Parsing Accuracy = $460 / 500 \times 100 = 92.0\%$

For Derivational words: Parsing Accuracy = $405 / 500 \times 100 = 81.0\%$

Therefore, Inflectional words show a parsing accuracy that is: $92.0\% - 81.0\% = 11.0$ percentage points higher

The relative difference is approximately: 13.58%

Statistical Test

A chi-square test of independence is used to determine whether correct parsing is associated with morpheme type.

Observed parsing outcome table:

Morpheme Type	Incorrectly Parsed	Correctly Parsed	Total
Inflectional	40	460	500
Derivational	95	405	500

The statistical analysis gives:

- Chi-square statistic ~ 25.906
- $p\text{-value} \sim 0.0000004$ ($p < 0.0001$)

Since $0.0000004 < 0.05$, the null hypothesis is rejected.

Therefore, the difference in rule-based parsing accuracy between inflectional and derivational word forms is statistically significant at the 5% level.

MORPHOLOGICAL PARSING - STATISTICAL TEST

Inflectional Accuracy: 92.0%

Derivational Accuracy: 81.0%

Accuracy Gap: 11.0 percentage points

Chi-square Statistic: 25.9060

p-value: 0.0000004

Significance Level: 0.05

RESULT: Statistically Significant

DECISION: Reject H0

RECOMMENDATION: Prioritize derivational-rule expansion

12. Data Visualization

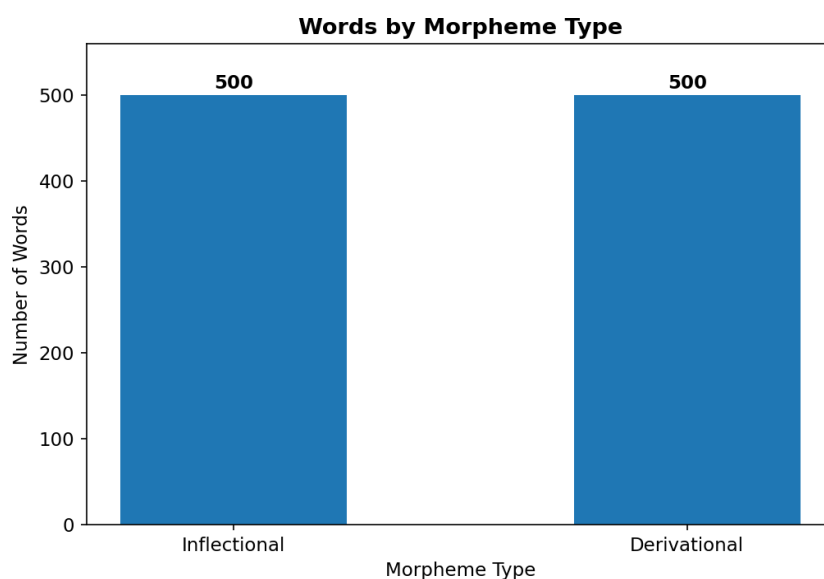
The following visualizations were produced for the project.

Visualization 1: Words by Morpheme Type

A bar chart showing:

- Inflectional -> 500
- Derivational -> 500

Purpose: To demonstrate balanced sample allocation.

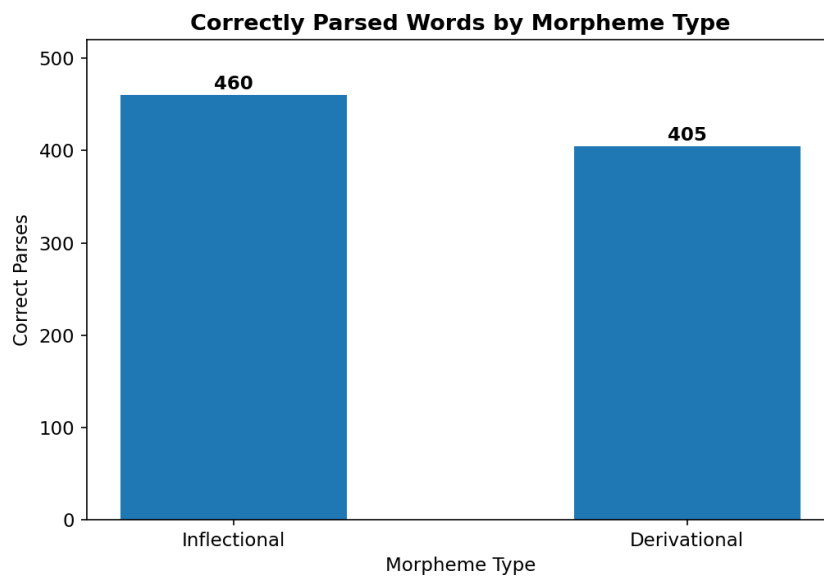


Visualization 2: Correctly Parsed Words by Morpheme Type

Bar chart:

- Inflectional -> 460 correctly parsed
- Derivational -> 405 correctly parsed

Purpose: To compare the absolute number of correct parses.

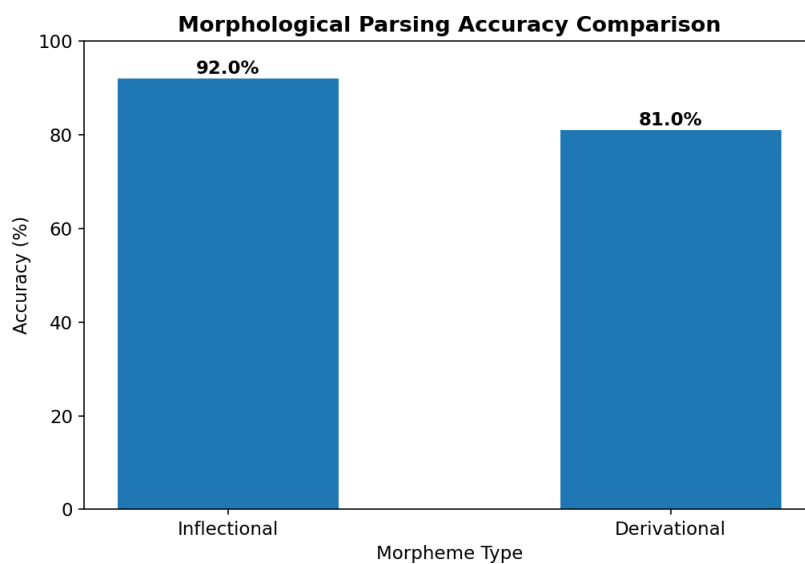


Visualization 3: Parsing Accuracy Comparison

Bar chart:

- Inflectional -> 92.0%
- Derivational -> 81.0%

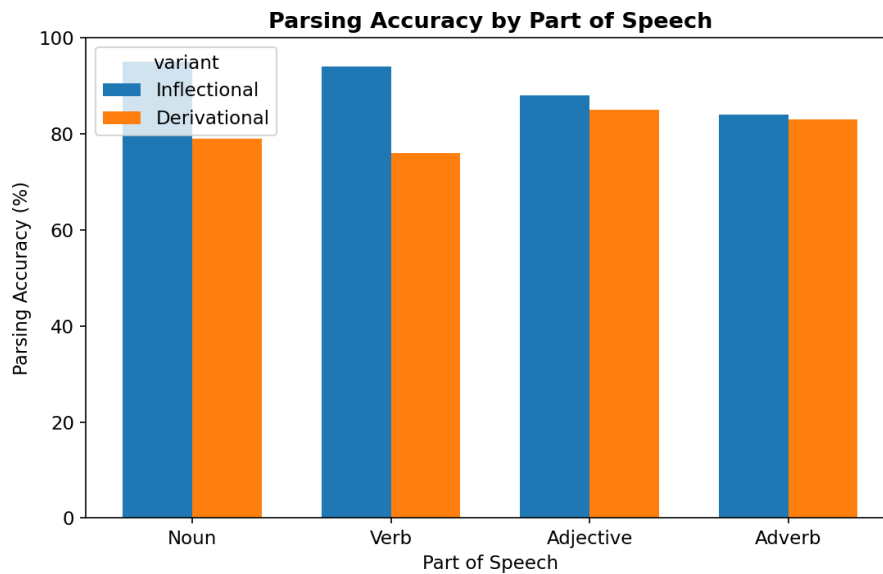
Purpose: To clearly identify the more accurately parsed morpheme category.



Visualization 4: Parsing Accuracy by Part of Speech

A grouped bar chart compares parsing accuracy for:

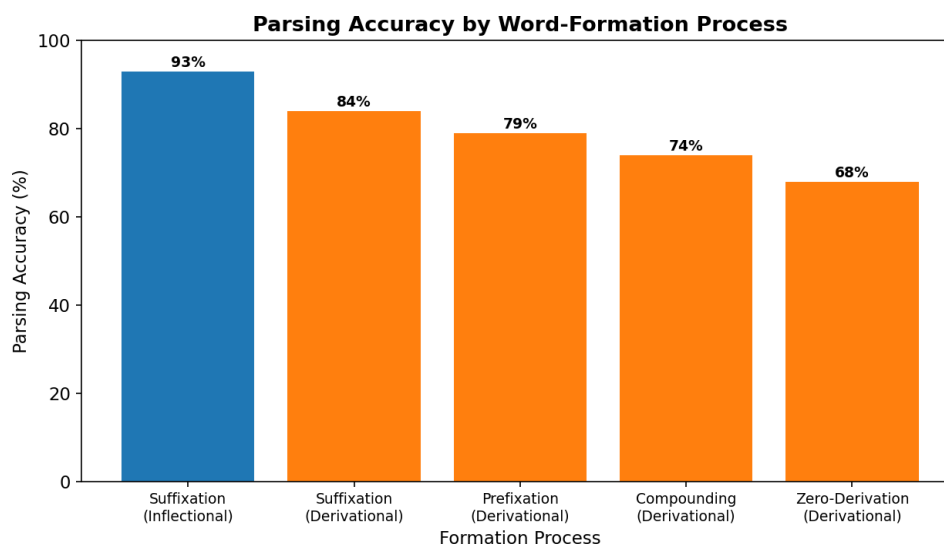
- Noun
- Verb
- Adjective
- Adverb



Visualization 5: Parsing Accuracy by Formation Process

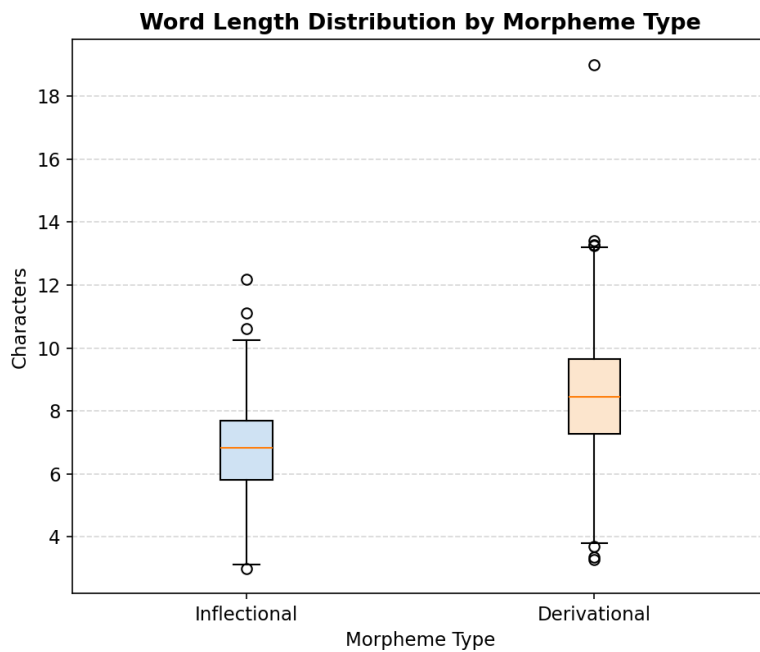
A bar chart compares parsing accuracy for:

- Suffixation (Inflectional)
- Suffixation (Derivational)
- Prefixation (Derivational)
- Compounding (Derivational)
- Zero-Derivation (Derivational)



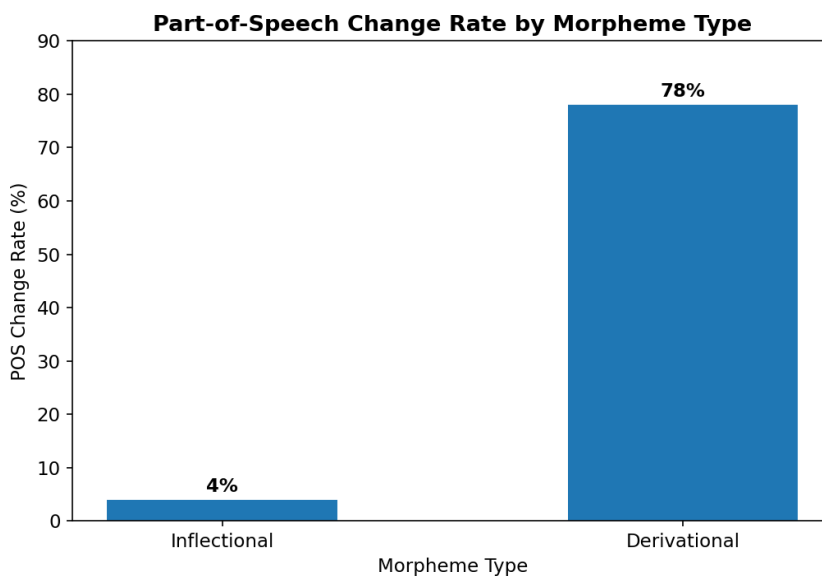
Visualization 6: Word Length Distribution

A box plot compares word length between Inflectional and Derivational forms.



Visualization 7: Part-of-Speech Change Rate by Morpheme Type

A bar chart compares how often each morpheme type changes the part of speech of the base word.



13. NLP Technique / Model Used

The primary NLP technique used is Rule-Based Morphological Analysis.

The project involves the following types of analytics:

Descriptive Analytics

Describes what happened by calculating:

- Number of words per morpheme type
- Number of correctly parsed words
- Parsing accuracy
- Average word length and syllable count
- Part-of-speech change rate

Diagnostic Analytics

Examines why one morpheme type may be harder to parse by analyzing:

- Part-of-speech category
- Formation process
- Word and affix length
- Whether the affix changes part of speech

Statistical Analysis

Hypothesis testing is used to determine whether the difference between the two morpheme categories is statistically significant.

Prescriptive Analytics

Based on the statistical findings, a recommendation is made regarding where the rule-based parser should be improved.

14. Model Development / Analysis

The morphological analysis follows these steps:

Step 1: Load Dataset

The CSV dataset of annotated words is imported using Pandas.

Step 2: Separate the Morpheme Categories

The data is divided into: Inflectional group and Derivational group.

Step 3: Apply the Rule-Based Morphological Analyzer

Each word is passed through a rule-based parser that strips known inflectional suffixes (-s, -es, -ed, -ing, -er, -est) and derivational affixes (prefixes such as un-, re-, dis-, pre- and suffixes such as -ness, -ment, -tion, -ize, -less, -ful, -ly) to recover the base word.

Step 4: Calculate Parsing Accuracy

Parsing accuracy is calculated separately for each morpheme category by comparing the parser's output against the gold-standard annotation.

Step 5: Formulate Hypotheses

H0: No difference exists in parsing accuracy. H1: A significant difference exists.

Step 6: Perform Statistical Test

A chi-square test is applied to the parsing-outcome results.

Step 7: Compare p-value with alpha

The calculated p-value is compared with 0.05.

Step 8: Draw Linguistic and NLP Conclusions

The morpheme category with lower parsing accuracy is identified as the priority area for rule expansion.

15. Model Evaluation / Validation

The rule-based analyzer is evaluated using:

- Parsing accuracy
- Accuracy difference between morpheme types
- Relative accuracy gap
- p-value
- Statistical significance
- Confidence interval

The estimated accuracy difference is: $92.0\% - 81.0\% = 11.0$ percentage points

The approximate 95% confidence interval for the difference is: 6.9% to 15.1%

Because the interval is above zero and the p-value is well below 0.05, the evidence strongly supports lower parsing accuracy for derivational word forms.

The analysis should still be interpreted in the context of the simulated academic dataset and its sample size.

16. Linguistic and NLP Insights

1. Inflectional words achieved a higher rule-based parsing accuracy than derivational words.
2. Inflectional forms achieved a 92.0% parsing accuracy.
3. Derivational forms achieved an 81.0% parsing accuracy.
4. Inflectional forms produced 460 correct parses compared with 405 for derivational forms.
5. Inflectional accuracy is 11.0 percentage points higher than derivational accuracy.
6. The chi-square test produced a p-value of approximately 0.0000004.
7. The observed accuracy gap is statistically significant at the 5% significance level.
8. Derivational words are, on average, longer and have more syllables than inflectional words.
9. Derivational affixes change the part of speech of the base word in about 78% of cases, compared with only 4% for inflectional affixes.
10. Compounding and zero-derivation formation processes show the lowest parsing accuracy overall.

17. Recommendations / Applications

Based on the analysis, the rule-based morphological analyzer performs reliably on inflectional forms but requires further development for derivational forms.

The NLP pipeline should consider the following improvements:

- Expand the derivational affix dictionary, particularly for compounding and zero-derivation cases.
- Introduce part-of-speech-aware rules, since derivational affixation frequently changes word category.
- Incorporate a statistical or neural morphological segmentation model alongside the rule-based system for low-accuracy categories.
- Continue monitoring parsing accuracy as new vocabulary is added to the system.

Further morphological analysis can also be conducted by examining:

- Irregular inflectional forms (e.g., go/went, child/children)
- Multiple-affix derivational chains (e.g., un- + help + -ful + -ness)
- Domain-specific vocabulary (technical, medical, legal terminology)

- Cross-linguistic morphological comparison

18. Implementation / Dashboard

The project can be implemented using Python.

Dashboard Components

A dashboard can contain the following sections:

KPI Cards

- Total Words Analyzed: 1,000
- Total Correct Parses: 865
- Overall Parsing Accuracy: 86.5%
- Higher-Accuracy Category: Inflectional

Comparison Section

- Inflectional Parsing Accuracy: 92.0%
- Derivational Parsing Accuracy: 81.0%
- Accuracy Gap: 11.0 percentage points

Statistical Section

- Test: Chi-square Test of Independence
- p-value: 0.0000004
- Significance Level: 0.05
- Result: Statistically Significant

Charts

1. Words by morpheme type
2. Correct parses by morpheme type
3. Parsing accuracy comparison
4. Accuracy by part of speech
5. Accuracy by formation process
6. Word length distribution
7. Part-of-speech change rate comparison

A Python dashboard can be implemented using Streamlit or Flask.

MORPHOLOGICAL PARSING PERFORMANCE

TOTAL WORDS	TOTAL CORRECT PARSES	INFLECTIONAL ACCURACY
1,000	865	92.0%

DERIVATIONAL ACCURACY	ACCURACY GAP	P-VALUE
81.0%	11.0 pts	0.0000004

RECOMMENDATION: EXPAND DERIVATIONAL RULE SET

Inflectional = 92.0% accuracy vs Derivational = 81.0% accuracy; $p < 0.05$

19. Results and Discussion

The morphological analysis experiment produced a clear difference between the two morpheme categories.

Inflectional word forms numbered 500 and produced 460 correct parses, resulting in a 92.0% parsing accuracy.

Derivational word forms also numbered 500 but produced 405 correct parses, resulting in an 81.0% parsing accuracy.

Therefore, the rule-based analyzer correctly parsed 55 more inflectional words than derivational words from the same sample size.

The accuracy gap was 11.0 percentage points, corresponding to an approximate relative difference of 13.58%.

The statistical test produced a p-value of approximately 0.0000004. Since the p-value is far less than the selected significance level of 0.05, the null hypothesis is rejected.

Thus, the analysis provides statistical evidence that rule-based parsing accuracy differs between the two morpheme categories, with inflectional forms being parsed more reliably in this dataset.

From an NLP-engineering perspective, derivational morphology is therefore the priority area for further rule development.

20. Limitations

1. The dataset is curated for academic purposes rather than drawn from a live production text corpus.
2. The sample contains only 1,000 word forms.
3. The experiment does not include multi-affix or nested morphological chains in depth.
4. Irregular and exceptional word forms are not fully represented.
5. The analysis focuses mainly on parsing accuracy as the primary success metric.
6. Syllable-count values are estimated and may not always match dictionary pronunciations exactly.
7. The results should not automatically be generalized to all English vocabulary or to other languages.
8. Real-world morphological analyzers should be evaluated on larger, domain-diverse corpora with proper cross-validation.

21. Future Enhancements

1. Using a real-world, large-scale English text corpus.
2. Increasing the sample size and vocabulary diversity.
3. Including irregular inflectional forms and multi-affix derivational chains.
4. Performing statistical or neural morphological segmentation (e.g., BPE, Morfessor).
5. Testing different affix-stripping rule sets and comparing performance.

6. Performing segmentation by word frequency and domain (technical, literary, conversational).
7. Applying machine-learning models to predict morpheme type and formation process.
8. Developing a real-time morphological-analysis dashboard.
9. Using probabilistic or Bayesian methods for ambiguous affix resolution.
10. Integrating the analyzer with standard NLP libraries such as NLTK or spaCy.
11. Performing cross-linguistic morphological comparison.
12. Automating rule-set monitoring and reporting as vocabulary grows.

22. Conclusion

This project demonstrated the use of computational morphological analysis to compare the performance of a rule-based parser on two categories of English word formation.

A dataset containing 1,000 word records was analyzed using exploratory data analysis and statistical testing. Inflectional word forms achieved a 92.0% parsing accuracy, while derivational word forms achieved an 81.0% parsing accuracy.

Inflectional forms therefore showed an 11.0 percentage-point higher accuracy and approximately 13.58% relative improvement compared with derivational forms.

The statistical analysis produced a p-value of approximately 0.0000004, which is far lower than the significance level of 0.05. Hence, the null hypothesis was rejected and the observed difference was considered statistically significant.

Based on the analysis, derivational morphology is recommended as the priority area for rule-set expansion in future NLP system development, subject to confirmation through evaluation on larger, real-world corpora.

23. Tools and Technologies Used

Tool / Technology	Purpose
Python	Data analysis, rule-based parsing, and statistical analysis
Pandas	Data loading, cleaning and manipulation
NumPy	Numerical operations
NLTK	Tokenization, POS tagging and morphological reference resources
Matplotlib	Data visualization
Seaborn	Statistical visualization
SciPy	Statistical hypothesis testing
Jupyter Notebook / Google Colab	Development environment
Streamlit / Flask	Optional dashboard development
CSV	Dataset storage
GitHub	Project version control and sharing

24. Project / GitHub Link

GitHub Repository:

[Paste your GitHub repository link here]

25. References

1. Python Documentation - Python programming and data analysis.
2. Pandas Documentation - Data manipulation and analysis.
3. NumPy Documentation - Numerical computing.
4. SciPy Documentation - Statistical analysis and hypothesis testing.
5. NLTK Documentation - Natural language toolkit for morphological and POS resources.

6. Matplotlib Documentation - Data visualization.
7. Seaborn Documentation - Statistical data visualization.
8. Standard linguistics references on English inflectional and derivational morphology.
9. Course materials for AL23531 - Natural Language Processing.